



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
FULL SYLLABUS
30-03-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE (ALL STAR BATCH)]

PAPER-2

Time : 3 Hours

Maximum Marks : 204

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

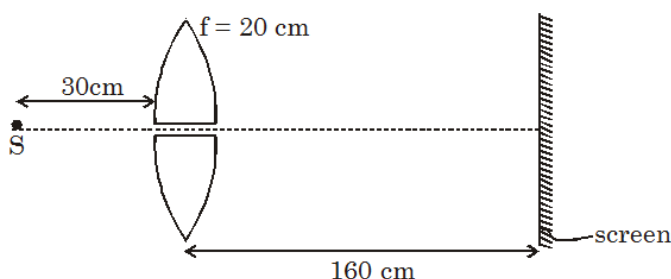
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. A person is riding a cycle at a speed of 18 km/hr along the equator. The wheel consists of 18 spokes and has a diameter of 50 cm. A bulb is attached between centre of wheel and its periphery. If the resistance of bulb is $0.3 \mu\Omega$, then power dissipated in the bulb will be (given $\vec{B}_{\text{earth}} = 25\mu\text{T}$ & assume no slipping) :-
 (A) 14.65 mW (B) 1.628 mW (C) 0.814 mW (D) 0 mW
2. A hydrogen atom and a He^+ atom are approaching each other with speed v & $2v$ respectively. The minimum kinetic energy that He^+ atom must have so that both the atom can excite to 1st excited state will be :-
 (A) 20.4 eV (B) 113.33 eV (C) 81.6 eV (D) 102 eV

Space for Rough Work

3. A point source of light 'S' is kept at the bottom of a tank that has a liquid filled to a height 2m. When seen from outside (air) and at normal incidence, the source of light appears to be raised by 40 cm. An opaque disc of radius $\frac{500}{3}$ cm is now placed on the surface of liquid. It is made of a material such that it floats on the surface of liquid with its centre always vertically above the source S. Starting at $t = 0$, radius 'r' of the disc is increased at a rate $\frac{dr}{dt} = 2t$ cm/sec. At a time t' (sec), it is found that the source can be just seen from outside. The radius of opaque disc is allowed to increase till a time $t' + 10$ (sec) and then no change subsequently. From the instant t' to $t' + 10$, the source could not be seen from outside. More liquid is now added to the tank and it is found that the source can just be seen from outside when the height of liquid in the tank becomes 'h'. Find the value of h
- (A) 4.25 m (B) 2.25 m (C) 5 m (D) 3.25 m
4. Diagram shows a modified YDSE set-up in which a lens is used for creating two coherent source by cutting the lens from middle & displacing them by 0.1 mm each in the direction perpendicular to the principle axis of the lens. If the source emits a light of wavelength $\lambda = 500$ nm, then the fringe width of the interference pattern formed on the screen will be :-



- (A) 833.33 μm (B) 203.33 μm (C) 238 μm (D) 630.33 μm

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

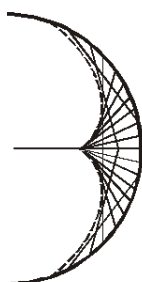
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. If the aperture of a spherical mirror is not very small, the reflected light doesn't meet at the focus even when they are incident parallel to the principle axis. This is referred as spherical observation. The reflected rays looks like they are bounded by an envelope known as caustic curve (shown in dashed line). Assuming center of the sphere to be origin & radius of curvature to be 'a'. Choose the correct option(s):

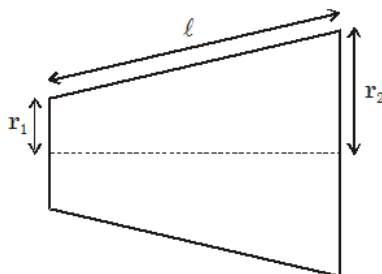


Incident rays are parallel to principle axis and is not shown in diagram

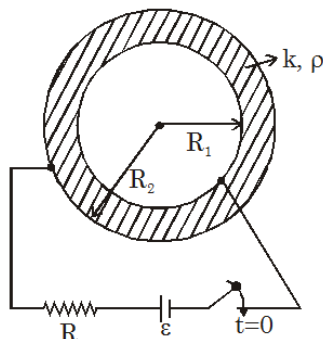
- (A) For a ray coming at height h , the equation of reflected ray is given by $y = \frac{h}{a^2 - 2h^2} [2\sqrt{a^2 - h^2}x - a^2]$
- (B) The ray in option 'A' will cross x-axis at a point such that its distance from origin is $\frac{a^2}{\sqrt{a^2 - h^2}}$
- (C) The equation of the caustic curve is given by $f(x, y, \theta) = x \tan 2\theta - \frac{a \sin \theta}{\cos 2\theta} - y = 0$
(θ is angle of incidence)
- (D) For $\theta = 60^\circ$, reflected ray will pass through the pole of mirror.

Space for Rough Work

6. Diagram shows a resistor in the form of a frustum having resistivity ρ_0 and dimension as shown in figure is connected to the battery of emf 'V' volt. At $t = 0$, switch was closed and the room temperature was T_0 . Temperature coefficient of resistance of frustum is zero and also neglect any change in dimension of frustum due to heating. Also assume all the heat produced in the frustum is utilized in heating the frustum only and consider frustum to be a black body. Choose the incorrect option(s) :



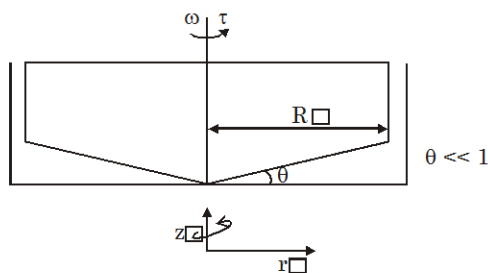
- (A) The temperature of the frustum will keep on increasing.
- (B) Maximum temperature achieved by frustum will be independent of heat capacity of frustum.
- (C) Maximum temperature achieved by the frustum is given by $\left[\frac{r_1 r_2 V^2}{\rho_0 \ell \sigma [r_1^2 + r_2^2 + \ell (r_1 + r_2)]} \right]^{1/4}$
- (D) On making the emf of battery $4V$, maximum temperature achieved by frustum will be 16 times as achieved when emf was V .
7. Figure shows concentric spheres of radius R_1 & R_2 respectively. The space between them is filled with dielectric of dielectric constant k & resistivity ρ . The switch was closed at $t = 0$. Choose the correct option(s) :



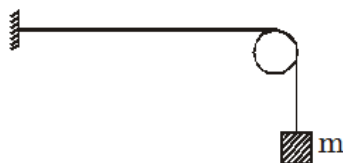
- (A) The value of maximum potential difference across inner and outer sphere will be ϵ .
- (B) Time constant for charging of RC circuit will be $\frac{4\pi\rho R\epsilon_0 k R_1 R_2}{[4\pi R R_1 R_2 + \rho (R_2 - R_1)]}$
- (C) In the steady state, current through R will be zero.
- (D) Maximum charge appearing on the inner & outer sphere is dependent of R .

Space for Rough Work

8. A common type of viscometer consists of a cone rotating against a fixed plate, as shown in figure. Choose the correct option(s) :-



- (A) The shear rate $\left(\frac{dv}{dh}\right)$ increases as we move outward from the axis of rotating cone.
- (B) Total torque needed to be applied so that ' ω ' remains constant is proportional to R^3
- (C) The ratio of total torque applied and ω achieved will be constant .
- (D) The dynamic co-efficient of viscosity of the fluid in between is given by $\frac{3\tau\theta}{2\pi R^3\omega}$
9. A string fixed at one end and supported by a weight at the other end (as shown in figure) is oscillating in 5th harmonic when a body of mass ' m ' is hung at the other end using an oscillator of frequency ' f '. Choose the correct option(s) :-



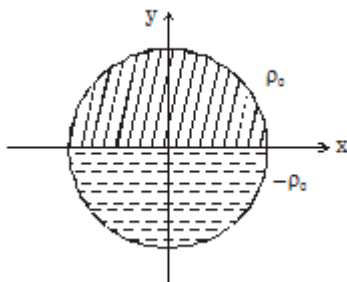
- (A) If we want the frequency of oscillator be fundamental, mass of block must be made 25 m.
- (B) For ' f ' to be fundamental frequency we can use thick string in place of the given string .
- (C) For ' f ' to be fundamental frequency we can submerge the block in liquid of density less than density of block.
- (D) On performing the same experiment on moon the string will not oscillate in any mode for given frequency & mass pair.

Space for Rough Work

10. Choose the correct statement(s) :-
- (A) Increasing accelerating potential in standard Coolidge tube will convert a soft x-ray into a hard x-ray.
 - (B) Presence of K_{α} x-ray ensure the presence of K_{β} x-ray too.
 - (C) If in a Coolidge tube, L_{α} lines are present then at least some K_{α} lines will also be present
 - (D) On increasing the filament voltage in Coolidge tube will increase the intensity of x-rays.
11. One mole of an ideal monoatomic gas is taken from state A to state B through the process $P = \frac{3}{2}T^{1/2}$. It is found that its temperature increases by 100 K in this process. Now it is taken from state B to C through a process for which internal energy is related to volume as $U = \frac{1}{2}V^{1/2}$. It is given that volume at B is 100 m^3 and at C it is 1600 m^3 . [Use $R = 8.3 \text{ J/mol-K}$]. Choose the correct option(s) :
- (A) Work done in process AB is 415 J
 - (B) Work done in process BC is 20 J
 - (C) Total work done is 415 J
 - (D) Total work done is 395 J

Space for Rough Work

12. A solid sphere of radius R , is centered at the origin. The northern hemisphere carries a uniform charge density ρ_0 , and the southern hemisphere a uniform charge density $-\rho_0$ as shown in figure. Choose the correct statement(s) :



- (A) The value of net dipole moment of the system is $\frac{\pi\rho_0 R^4}{2} \hat{j}$.
- (B) The value of electric potential at a point P which is at a distance ' r ' ($\gg R$) from the origin and $\vec{r}$ makes an angle θ with x-axis is given by $\frac{\rho_0 R^4 \cos \theta}{8\epsilon_0 r^2}$
- (C) Value of electric field at the point Q on x-axis at a distance r ($\gg R$) is $\frac{\rho_0 R^4}{8\epsilon_0 r^3} (-\hat{j})$
- (D) If an external electric field $\vec{E} = -E_0 \hat{j}$ is applied & the sphere is slightly rotated about z axis, the maximum kinetic energy achieved by the sphere is $\frac{\pi\rho_0 R^4 E}{4}$

Space for Rough Work

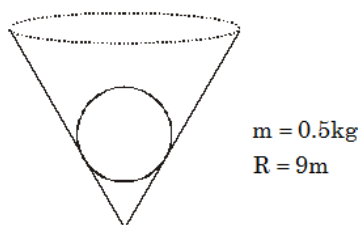
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

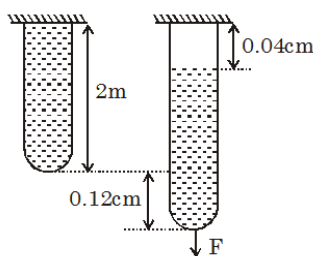
Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

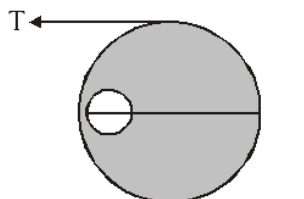
1. A hollow sphere (mass m , radius R) is put inside a hollow cone (mass m , radius $5R$ and semi-vertex angle 37°) as shown in the figure. The whole arrangement is rigidly fixed to the ground at the apex of the cone. Both the sphere and the cone are made of the same material ($\alpha = 10^{-4} / ^\circ\text{C}$). Considering that major changes in dimensions could happen only due to heating effects, find the increase in potential energy (in Joule) of the system if its temperature is increased by 100°C .



2. A uniform glass tube of length 2m is closed at the lower end. It is completely filled with water and rigidly fixed at upper end. Now the tube is stretched downwards. Although the length of tube increases by 0.12 cm , the length of water column increases only by 0.08 cm . Assume water is incompressible. What is poisson's ratio σ for glass ?

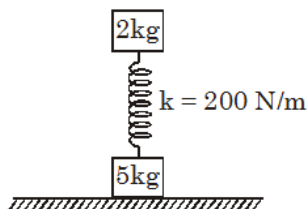


3. A uniform cylinder of radius ' a ' originally had a weight of 80 N . After an off-axis cylindrical hole at $2a/3$ was drilled through it as shown, it weighs 65 N . The axes of the two cylinders are parallel and their centres are at the same height. A force T is applied to the top of the cylinder horizontally. Find the value of the force (in N) required in order to keep the cylinder at rest.



Space for Rough Work

4. The radius of curvature of a meter gauge railway line at a place where train is moving at 54km/hr is 150m. If there is no side thrust on the rails, the elevation of the outer rail above the inner rail is given by α cm. Find the value of α .
5. Figure shows two blocks connected to a spring of spring constant 200 N/m. Initially 2kg block is in equilibrium. Now if 2kg block is further pushed downward by 35 cm and is released, 5kg block leaves contact after ' α ' (sec) from the surface. Find the value of ' α '.



6. A block of mass 2 kg is released from point Q on a frictionless track which ends in quarter-circular track of radius 5m at the bottom as shown. Find the magnitude of total acceleration (in m/s^2) of the block when it arrives at P.



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. pH of a 0.1M solution of HCl is changed by 0.3 units due to dilution, calculate change in osmotic pressure if temperature of solution is 300K. ($R = 1/12$ litre atm/K/mole) [$\log 2 = 0.3$]

(A) 2.5 atm (B) 5 atm (C) 7.5 atm (D) 1.25 atm

- 2.** Following are the examples of silicate minerals

Zircon, ZrSiO_4 Beryl, $\text{Be}_3\text{Al}_2\text{Si}_6\text{O}_{18}$ Pyrophyllite, $\text{Al}_2(\text{OH})_2[(\text{Si}_2\text{O}_5)_2]$

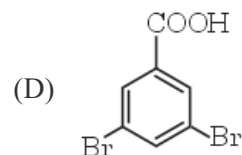
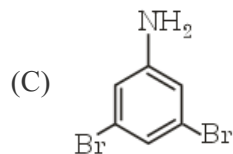
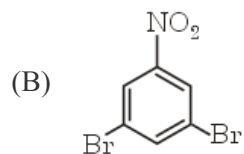
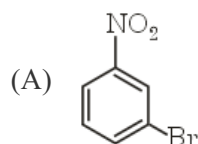
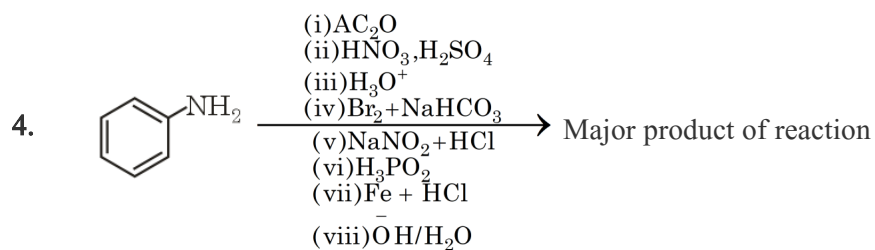
The correct structural description of the minerals is

- (A) **I**-Ortho silicate, **II**-Cyclic silicate and **III**-Sheet silicate
(B) **I**-Ortho silicate, **II**-Sheet silicate and **III**-Cyclic silicate
(C) **I**-Cyclic silicate, **II**-Sheet silicate and **III**-Ortho silicate
(D) **I**-Sheet silicate, **II**-Ortho silicate and **III**-Cyclic silicate

Space for Rough Work

3. The ratio of initial activities of two radioactive nuclides α and β in mixture was 8 : 1. After 2 hours ratio of their activities became 1 : 1. If $t_{1/2}$ of α nuclide is 20 minutes, then $t_{1/2}$ of β nuclide is.

(A) 40 min. (B) 20 min. (C) 80 min. (D) 120 min.



Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

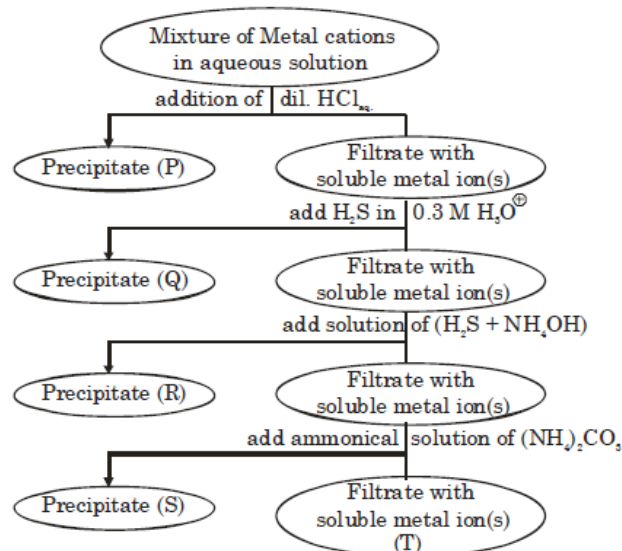
5. Which is/are similar for Cr & Mn :-
- (A) Both have 13 electrons in (n-1) shell
- (B) Both have tetrahedral coloured oxo anion in their highest oxidation state
- (C) MCl_2 salt of both are oxidised to highest oxidation state of metal on fusion with Na_2CO_3 & air
- (D) Both transition metals have $E_{\text{M}^{+2}/\text{M}}^0$ have +ve value
6. Select correct statement(s) for colloids -
- (A) For micellar sols there is a transition region corresponding to a range of concentrations around the CMC where physical properties vary smoothly but non-linearly with the concentration.
- (B) At temperatures below Kraft temperature if concentration of ionic surfactants in water is greater than CMC, they form an insoluble solid
- (C) Congo red sol is negative sol
- (D) Methylene blue sol is positive sol

Space for Rough Work

7. An aqueous solution initially has 0.1 M of salt M_2X in it. If $K_b(MOH) = 2 \times 10^{-5}$, $K_{a1}(H_2X) = 10^{-6}$, $K_{a2}(H_2X) = 5 \times 10^{-10}$. Identify correct option(s) at equilibrium.

- (A) $[X^{2-}] \approx 0.033 \text{ M}$ (B) $\text{pH} \approx 9$
(C) $[M^+] \approx 0.067 \text{ M}$ (D) $[HX^-] \approx 0.067 \text{ M}$

8.

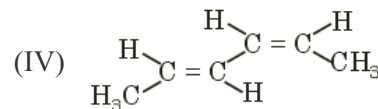
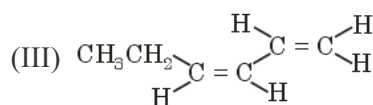
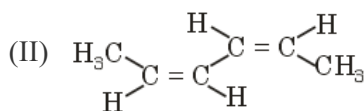
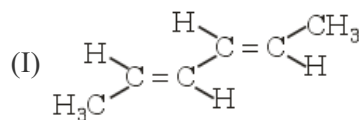


In above reaction sequence (P) (Q) (R) (S) and (T) can be respectively :-

- (A) Hg_2Cl_2 , CuS , Cr_2S_3 , $BaCO_3$, K^+ (B) $AgCl$, Bi_2S_3 , MnS , $CaCO_3$, Na^+
(C) $HgCl_2$, CdS , CoS , $SrCO_3$, Mg^+ (D) $PbCl_2$, HgS , $Al(OH)_3$, K_2CO_3 , Na^+
9. What is the sequence of amino acids in the peptide containing gly, lys, val, ala, that yields the following peptide fragments on partial hydrolysis? ala-val, lys-gly, gly-ala and val-gly
- (A) gly-gly-lys-val-ala (B) ala-val-gly-lys-gly
(C) lys-gly-ala-val-gly (D) val-gly-ala-lys-gly

Space for Rough Work

10. Arrange the following isomers of hexadiene in order of increasing stability (least stable first)



- (A) II, IV, III, I (B) III, II, IV, I (C) I, IV, II, III (D) III, II, I, IV
11. A pale blue diamagnetic gas (Y) is best prepared by flowing paramagnetic gas (X) at 1 atm and 25°C through concentric metallized glass tubes to which low frequency power at 50-500 Hz and 10-20 kV is applied to maintain a silent electric discharge.

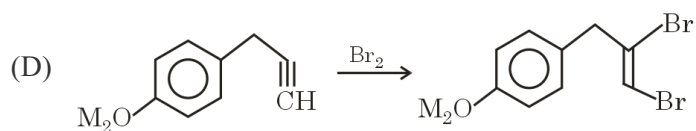
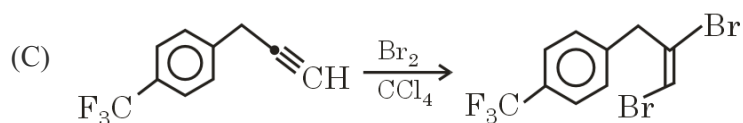
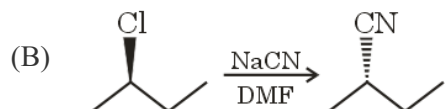
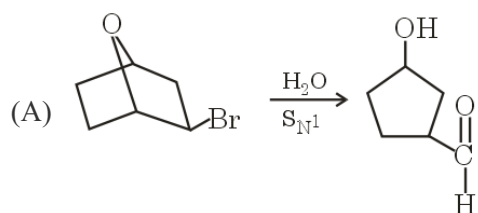
Gas (X) is colourless in gas form but liquid is pale blue. Gas (Y) is used for sterilization of foodstuffs and disinfection, and also form during the generation of photochemical smog. (X) & (Y) are the different form of a element which is about 47% by mass of earth crust.

Identify correct statement(s) regarding (X) & (Y)

- (A) Concentration of 'Y' in the mixture of X & Y is determined by iodometrically using H_3BO_3 & $\text{Na}[\text{B}(\text{OH})_4]$ buffer.
- (B) Conversion of (X) to (Y) is an exothermic process
- (C) 'Y' can oxidise black lead sulphide
- (D) Y is used to convert potassium manganate to potassium permanganate

Space for Rough Work

12. Identify reactions correctly matched with then major product?



Space for Rough Work

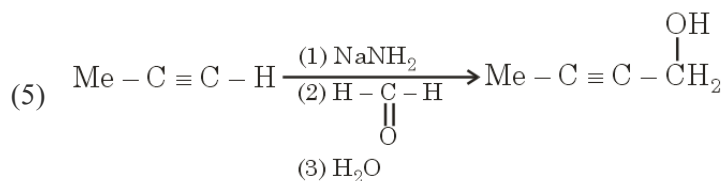
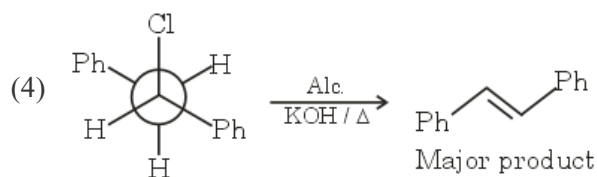
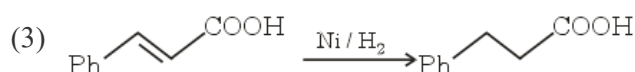
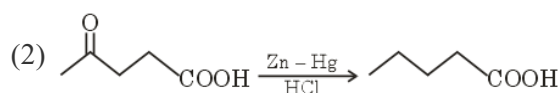
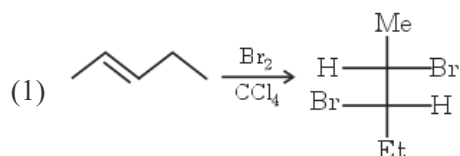
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Identify total number of reactions matched with their major products?

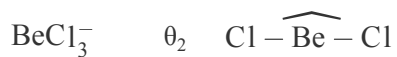
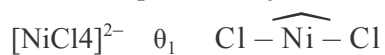


Space for Rough Work

2. The existence of coordination compounds with the same formula but different arrangements (Isomers) is crucial in the development of coordination chemistry. There are two main forms of isomerism in coordination compound structural isomerism and stereoisomerism (spatial isomerism). Find total possible isomeric compound for complex ion $[\text{Pt}(\text{Br})(\text{NO}_2)(\text{NO}_3)(\text{SCN})]^{2-}$
3. Identify total number of condensation co-polymer?
(1) Polyethylene (2) Tafflon (3) Cane sugar (4) Starch
(5) Cellulose (6) Bakalite (7) Glyptal
4. At 298 K, bond dissociation enthalpies of C–C and C=C are 330 and 540 KJ/mol, respectively. If the enthalpy of polymerisation per mole of polyethene $[(\text{C}_2\text{H}_4)_n(\text{g})]$ from ethene gas at 298 K is – 6MJ, the value of 'n' is :-
-

Space for Rough Work

5. Consider specified adjacent bond angle in given molecule(s)/ions and find $(\theta_1 + \theta_2 + \theta_3 + \theta_4)$



6. Root mean square speed of an unknown gas at 727°C is 10^5 cm/second. Calculate molar mass of unknown gas (in gram/mole) [Take $R = \frac{25}{3}$ J/mole-K].

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Two perpendicular tangents are drawn from a point to ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$. If the chords of contact of the ellipse are tangents to a family of concentric circles, having the centres same as that of the ellipse, then ratio of maximum and minimum values of radius of concentric circles.
 (A) 2 (B) $\frac{9}{4}$ (C) $\frac{5}{2}$ (D) 5
2. Let two conics be given by P: $y^2 = 5x$ and H: $x^2 - y^2 = 6$, P and H intersect at points A and B in first and fourth quadrant respectively. Tangents to P and H at point 'A' intersect the x-axis at points L and M respectively then ratio of areas of $\triangle LAB$ and $\triangle MAB$ is
 (A) $\frac{12}{5}$ (B) $\frac{5}{12}$ (C) $\frac{12}{7}$ (D) $\frac{7}{12}$
3. The minimum number of real roots of $(x^2 + 3x + a)(x^2 + ax + 1) = 0$, is (where $a \in \mathbb{R}$)
 (A) 4 (B) 2 (C) 0 (D) 1
4. If $a_1, a_2, a_3, \dots, a_n$ are distinct positive numbers in G.P. and $x_1, x_2, x_3, \dots, x_n$ are positive numbers, such that $a_1^{1/x_1} = a_2^{1/x_2} = a_3^{1/x_3} = \dots = a_n^{1/x_n}$, then $x_1, x_2, x_3, \dots, x_n$ are always in
 (A) A. P. (B) G. P. (C) H. P. (D) A. G. P.

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. An exam consist of 11 problems selected randomly from a collection of 1000 different problems. For a student to pass, he needs to solve correctly at least six of the 11 problems . if the students knows to solve exactly half of 1000 problems, then the probability that the student pass the exam is

(A) $\frac{1}{3}$ (B) $\frac{1}{4}$ (C) $\frac{1}{5}$ (D) $\frac{1}{2}$

6. Let $z_1, z_2, z_3, \dots, z_{10}$ are roots of equation $z^{12} + z^{10} + z^9 + z^8 + z^7 + z^6 + z^5 + z^4 + z^3 + z^2 + 1 = 0$ such that $z_r^3 + 1 \neq 0 \forall r \in \{1, 2, 3, \dots, 10\}$, then value of $(z_1^{2021} + z_2^{2021} + \dots + z_{10}^{2021})$ is-

(A) 4 (B) 5 (C) -1 (D) 0

7. Sum of common roots of the equation $z^3 + 2z^2 + 2z + 1 = 0$ and $z^{1985} + z^{100} + 1 = 0$ is

(A) -1 (B) 1 (C) 0 (D) 1

Space for Rough Work

8. Using five basic colours (Red, Blue, Green, Black, Pink), different colours are made by mixing these two or more colours in equal proportion. The n vertical stripes are painted by all colours available such that no two consecutive stripes have same colour. then number of ways of painting 'n' stripes such that only alternate stripes are painted by a basic colour, is-
- (A) $2(130)^{n/2}$, If n is odd (B) $31(130)^{\frac{n-1}{2}}$, If n is even
 (C) $31(130)^{\frac{n-1}{2}}$, If n is odd (D) $2(130)^{n/2}$, If n is even
9. Let $A = [a_{ij}]_{n \times n}$ be a square matrix such that $a_{ij} = (3i + 2)5^j$. then the value of $\lim_{n \rightarrow \infty} (\text{trace}(A^n))^{1/n}$ is, ($n \in \mathbb{N}$)
- (A) $\frac{23}{16}$ (B) $\frac{16}{23}$ (C) $\frac{23}{32}$ (D) $\frac{32}{23}$
10. Let $y = \sum_{r=0}^{n-2} \frac{2r^2 + 6r + 5 - n(r+1)}{(n-r-1)^n C_{r+1}}$, $n \in \mathbb{N}$, $n \geq 2$ then
- (A) $y = n + \frac{1}{n}$ (B) $y = n - \frac{1}{n}$ (C) $y > 1$ (D) $y > 0$

Space for Rough Work

11. A, B, C are angles of a ΔABC , then the value of $\lim_{n \rightarrow \infty} \sum_{k=1}^n \begin{vmatrix} \sin A & \sin B & \sin C \\ \sin(A + kx) & \sin(B + kx) & \sin(C + kx) \\ \cos(A + kx) & \cos(B + kx) & \cos(C + kx) \end{vmatrix}$,
 $x \in \mathbb{R}$ is

(A) a positive integer

(B) a Whole number

(C) a Rational number

(D) an irrational number

12. If $g(x) = f(x) \left(\int_0^x f(t) dt \right)^{2021}$ is decreasing function and $f(x)$ is continuous in $\mathbb{R}$. then $f(2021)$ is-

(A) a natural number

(B) an integer

(C) an irrational number

(D) a rational number

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. If $\int (4\cot^3 x + \cot^2 x + \cot x - 2) e^x dx = (-e^x)(a \cot^2 x + b \cot x + 5) + c$, then value of $\frac{b}{2a}$ is, (c is constant of integration)
2. If $\int_0^{2021} f(x) dx = \frac{3}{4}$ & $\sum_{r=1}^{2021} \int_0^1 f(r-1+x) dx = K$ then the value of 'K' is-
3. If $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^n \sqrt[n]{7^i 5^{n-i}} = \frac{2}{(\ln a - \ln b)}$ then value of $\frac{a}{b}$ is

Space for Rough Work

4. $|\vec{a}| = |\vec{b}| = |\vec{c}| = |\vec{a} + \vec{b}| = 1$. $\vec{a} \cdot \vec{c} = 0$
 $\vec{a} = \frac{\hat{i} + \hat{j}}{\sqrt{2}}$, If for some $\vec{c}$, $|\vec{b} \times \vec{c}| = |\vec{a} \times \vec{c}|$ and $\vec{c} = p\hat{i} + q\hat{j} + r\hat{k}$,
 then value of $\left(\frac{p^4 + q^4 + r^4}{2}\right)$ is
 (here $\vec{a}$, $\hat{k}$ and $\vec{b}$ are linearly dependent vectors).
5. The equation of plane which passes through the line of intersection of the planes $x + y + z - 1 = 0$ and $x - y + z - 3 = 0$ and which is at maximum distance from the point $(3, 4, 5)$ is $ax + by + cz = d$ then the value of $\frac{a + b - c}{4d}$ is
6. The triangle ABC, right angled at 'C', has Median AD, BE and CF. AD lies along the line $y = x + 3$, BE lies along the line $y = 2x + 4$. If the length of the hypotenuse is 60 and area of $\Delta ABC = 1600K$ then value of 'k' is-

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Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

Space for Rough Work